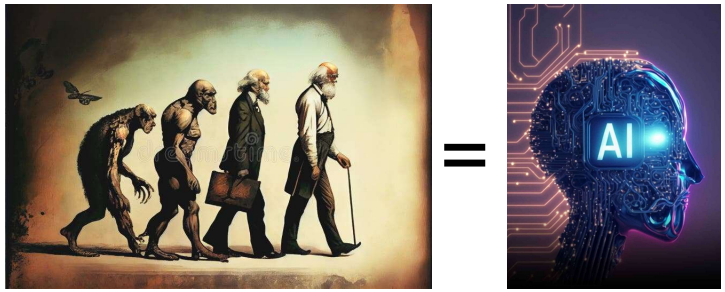
Validating AI data collection against traditional manual data collection methods



Tony Beatton[1,2] Steve Bickley[1] Ben Chan[1] Sophie Coulon[1] & Benno Torgler[1]

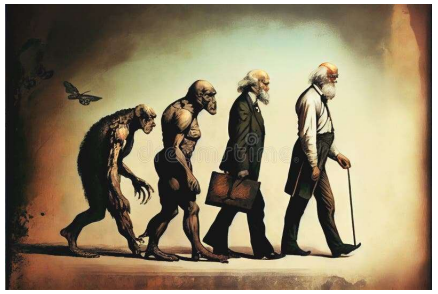
[1] ARC BITA, Queensland University of Technology
[2] Adjunct, University of Canberra

University of Canberra
Thursday 20th August 2026

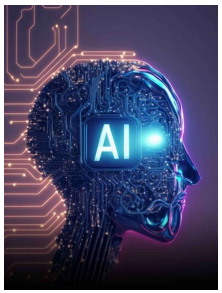
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Motivation:



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- **High cost** of manually collecting data with surveys;
- Collecting manual data are **slow.**

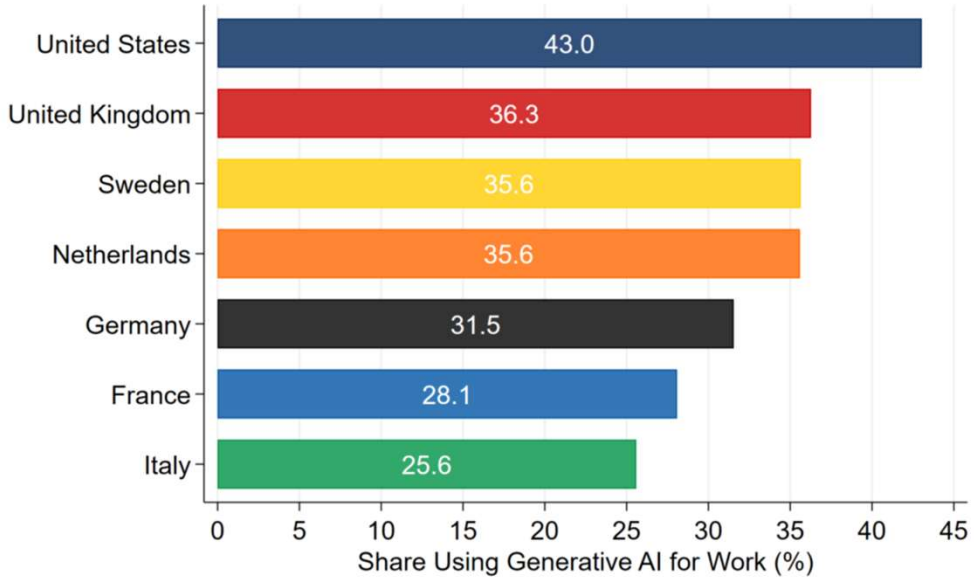
- **Develop & validate 'trusted' benchmark method(s)** that:
  - **Contributes to our trust in AI collected data.**
  - **Reduce project time and cost** of data collection and cleaning (~60%)
    - **Frees up time** for analyses and interpretation: **more thinking time**.
    - **Frees up funds** for additional research

Literature

- In Australia, Hegarty et. al., 2025 found **45.6% of Australian individuals have recently used a generative AI tool**
  - "*In the short time since OpenAI launched ChatGPT in November 2022, generative* ***artificial intelligence (AI) products have become increasingly ubiquitous and advanced*** "

- 'Around **12 per cent of Australian businesses** reported the use of AI in their workplace in 2024–25
  - "Around 35 per cent of large businesses reported using AI, up from 9 per cent in 2021–22, while 22 per cent of medium-sized businesses had adopted AI, compared with 3 per cent previously" (ABS 25/6/2026)

The share of survey respondents who use Generative AI for work



Notes: Data source: Authors' own survey of worker and firms run in January-February 2026. N = 20, 916 (Bick et. al., 2026)

Literature: Social-science research relies on gathering survey data for experiments.

- "*This **20-country survey of publishing academics** shows the **widespread awareness and adoption of generative AI tools in academia***" (Mohammadi et.al., 2026, p.1)

- "*Survey-taking AI tools surpass human abilities….*" (Panizza et. al., 2026, p. 293)
  - ***"Studies suggest that between 30% and 90% of responses to social-science surveys can be inauthentic or fraudulent."** (p. 293)*
  - ***AI-agent detection has been described as a continual game of cat and mouse, in which "the mouse never sleeps"….., we lay out four steps to minimize the risk of survey pollution by AI agents."** (p, 294)*

- *"**AI helps in minimizing human error**, improving data quality, and providing predictive insights that guide decision- making. Automated data collection through AI- powered tools thus reducing human bias and errors while speeding up the process."* (Dahiya et. al., 2026, p. 459)

Literature: Social-science research relies on gathering survey data for experiments.

- *"The increasing use of crowdsourcing platforms for **behavioural research rests on the assumption that research participants are exclusively human**. This **assumption is now under threat**. AI agents from browsers such as OpenAI's Atlas and Perplexity's Comet can autonomously complete online surveys. These agents can simulate specific personas or demographic profiles and follow survey prompts, select responses and submit data with fluency and internal consistency. Such capabilities threaten data authenticity and integrity, especially as subjective perception, motivation and emotion are central in behavioural research." (Thaichon et. Al., 2026)*

Literature: Study examining Survey Cost Reduction

Coffey & Elliot, 2024; **N = 8000**; **treatment: 550 optimization cases**

Coffey & Elliot, 2024 was supported by a grant from the National Institutes for Health (# 1R01AG058599-01).

ChatGP:

The average grant value from the National Institutes of Health (NIH) varies depending on the type of grant. For example, in fiscal year 2022, the average cost per Research Project Grant (RPG) was approximately $592,617[1].

Total Cost Comparison Coffey & Elliot, 2024 with SurveyLM/SEM

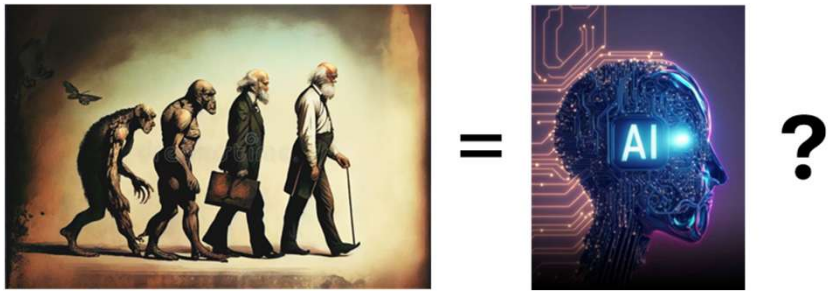
Coffey & Elliot, 2024:

NIH Estimation: $US592,617/8000 = $US74,08; **$AUD107.41 per observation**

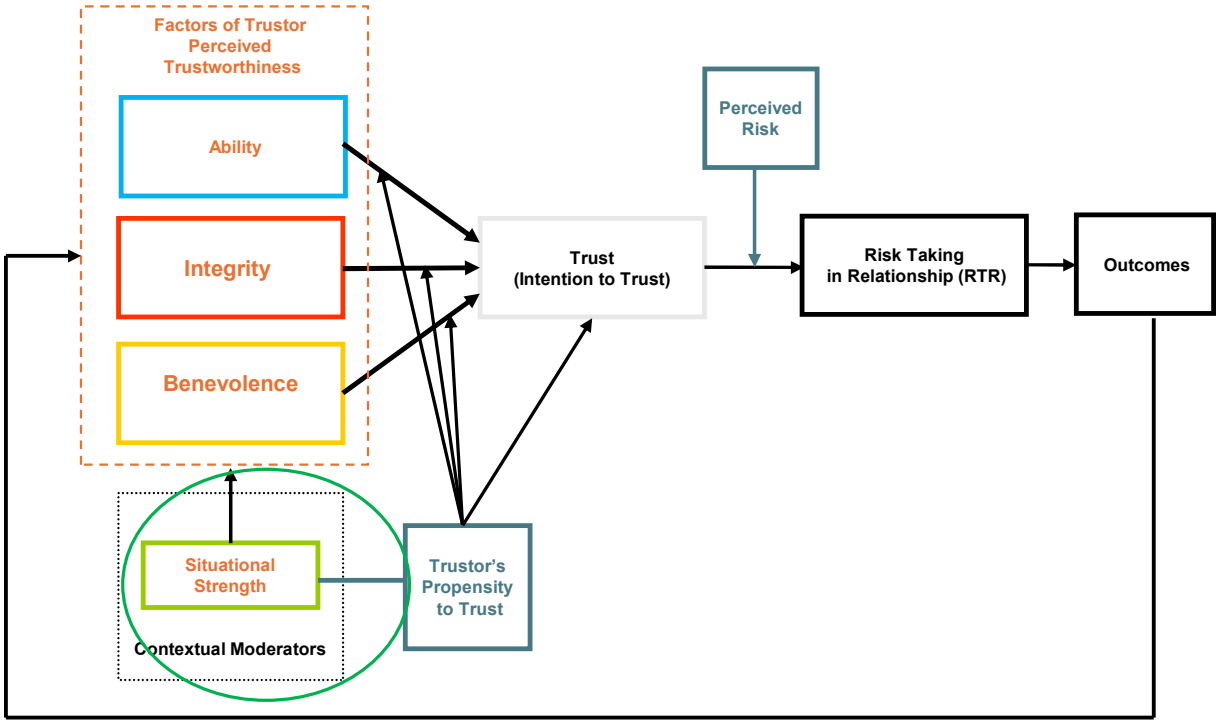
Based on Table 6:

	Treatment	Control
Data Collection	$311,170.00	$343,070.00
Estimates Salary I	$122,293.00	$123,743.00
Estimates Salary II	$121,916.00	$122,162.50
Total $AUD:	$555,379.00	$588,975.50
Cost per Obs	**$ 69.42**	**$ 73.62**

Will the AI collected data be theoretically & statistically equivalent to Manual Methods?



Develop & Validate a Trusted Benchmark Method



Adapted from Schoorman, Mayer & Davis, 1996

$$ITT = \beta_1(A * SS) + \beta_2(I * SS) + \beta_3(B * SS) + \epsilon$$

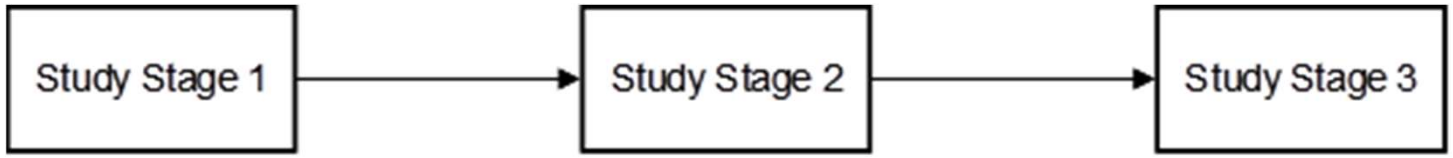
Treatment: Situational Strength

	Ability		Benevolence		Integrity	
Situational Strength (Moderator)	High	Low	High	Low	High	Low
Strong	✓		✓		✓	
Ambiguous		✓	✓			✓
Weak				✓		

Example dimensions of Situational Strength;
adapted from (Gill et al., 2005)

3x3 Factorial Design

Manual Data Collection



Vignette Development
- Qualitative
- Structured Interview
- Thematic analysis
- n = 20
- Vignette content
- Effect size

Pilot
- Quantitative
- Questionnaire
- Manipulation questionnaire
- n = 187
- Refine vignette & measures

Final Data Collection
- Quantitative
- Questionnaire
- Manipulation questionnaire
- n = 224
- Data analysis

Were the AI collected data be theoretically & statistically equivalent to Manual Methods?



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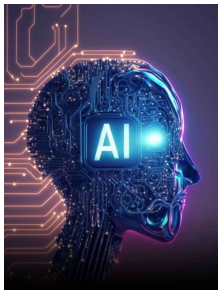


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Will the SurveyLM collected data be statistically equivalent to Manual Methods?



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Research Question (Ha): The Manual versus SurveyLM/AI collected data items are equivalent.

AI - Manual Comparison Method

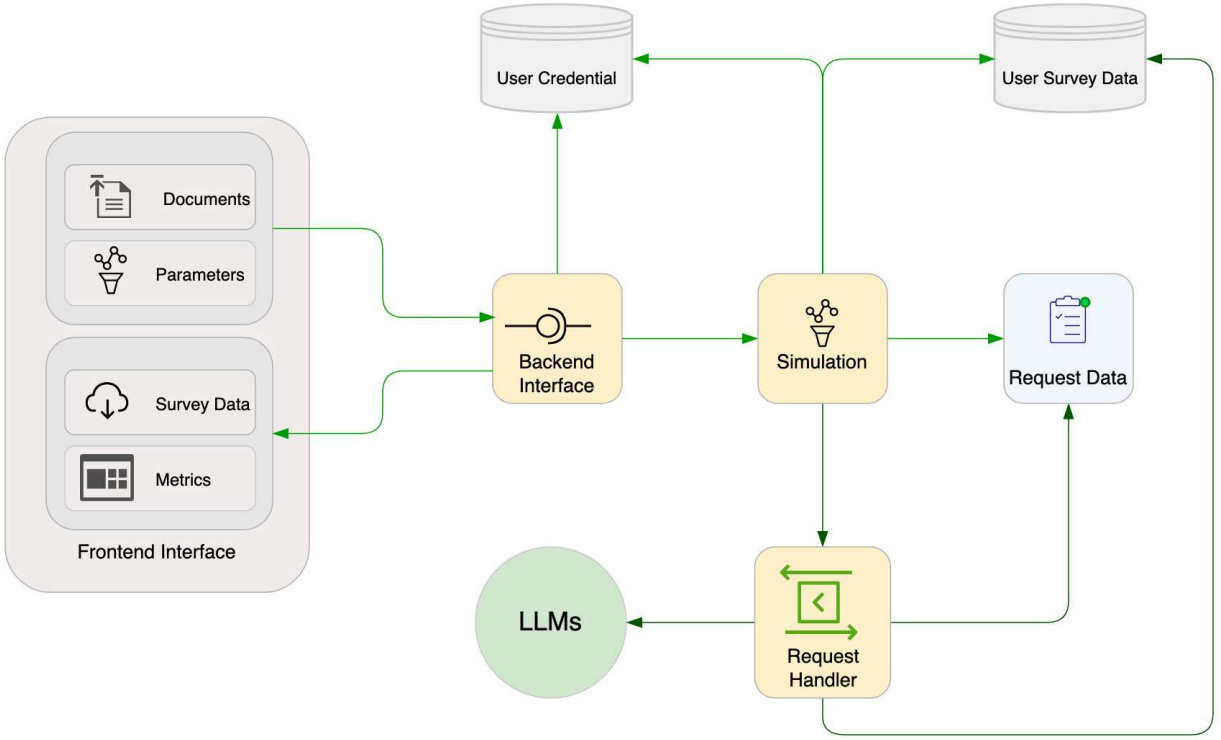
1. Use SurveyLM api connected to OpenAI Model(s) to collect data
   - Convert vignettes to text documents suitable for SurveyLM/OpenAI
   - Collect data
2. Test Manual – AI data equivalence
   - *Descriptive Statistics; Cronbach's Alpha; factor analysis;*
   - *tests of distributional equivalence: two sample t-tests*; Chi square; *Kolmogorov-Smirmov*
3. Test of Theory (Schoorman, Mayer, & Davis, 1996)
   - *SEM*

1. SurveyLM api & OpenAI LLMs to collect data

Documents from Manual Survey:

- **Survey questions**
- Vignettes that explain Situational Strength
- **Agent definitions:** demographics of the subjects who answered the manual surveys

Analysis to determine whether Manual and **SurveyLM Data** are identical.



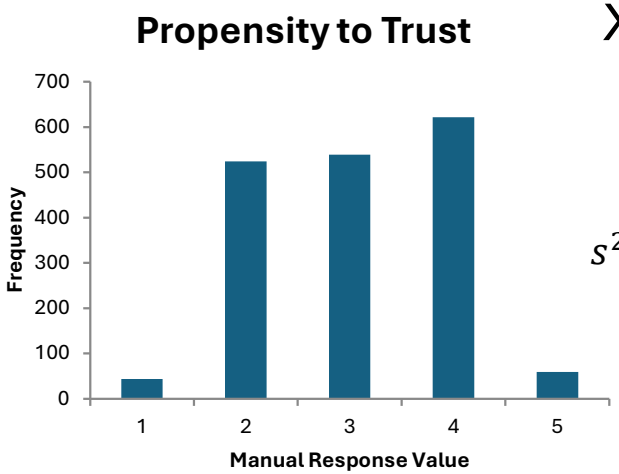
Panalogy Lab

JSON

2. Test Manual – AI data equivalence

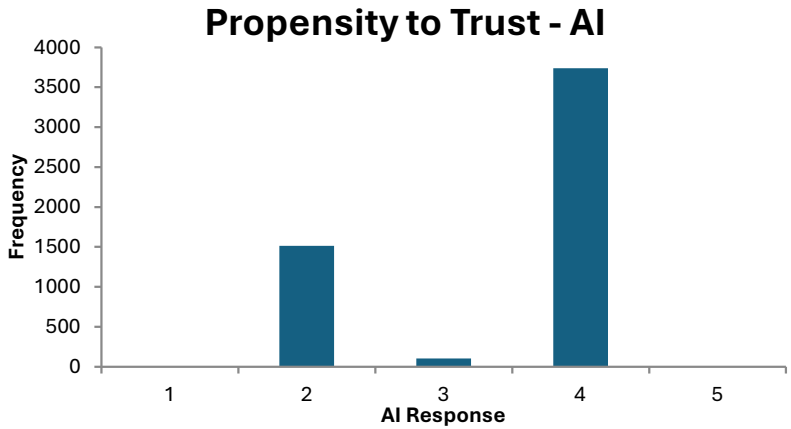
- Descriptive Statistics; Cronbach's Alpha; factor analysis;
- **tests of distributional equivalence**: two sample t-tests; Chi square; Kolmogorov-Smirmov

Results



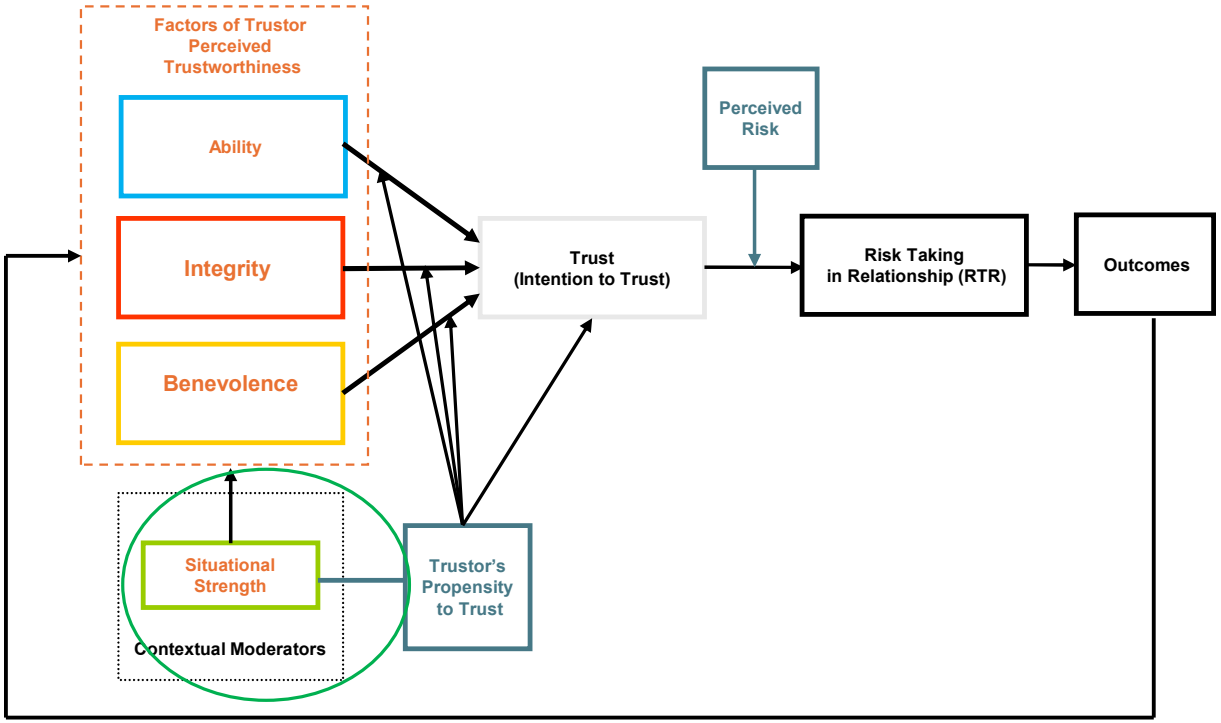
$$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$ (3.07; 3.42)

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$ (0.93; 0.90)



$$r_{(x,y)} = \frac{Cov(x,y)}{S_x * S_y}$$

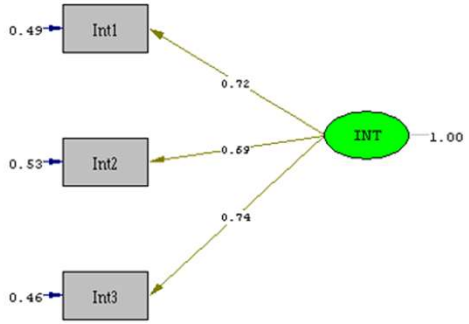
Develop & Validate a Trusted Benchmark Method



Adapted from Schoorman, Mayer & Davis, 1996

$$ITT = \beta_1(A * SS) + \beta_2(I * SS) + \beta_3(B * SS) + \epsilon$$

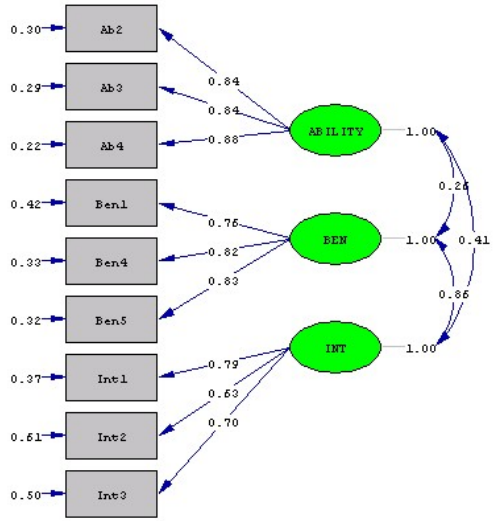
3. Test of Theory (Schoorman, Mayer, & Davis, 1996)



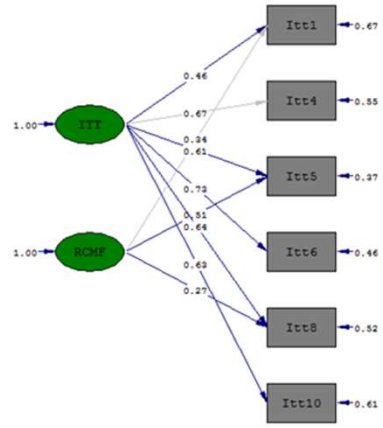
Chi-Square=0.65, df=2, P-value=0.72168, RMSEA=0.000

$$Cov(X,Y) = \frac{\sum(x-\bar{x})(y-\bar{y})}{n-1}$$

Standard error (e) = $\frac{s}{\sqrt{n}}$

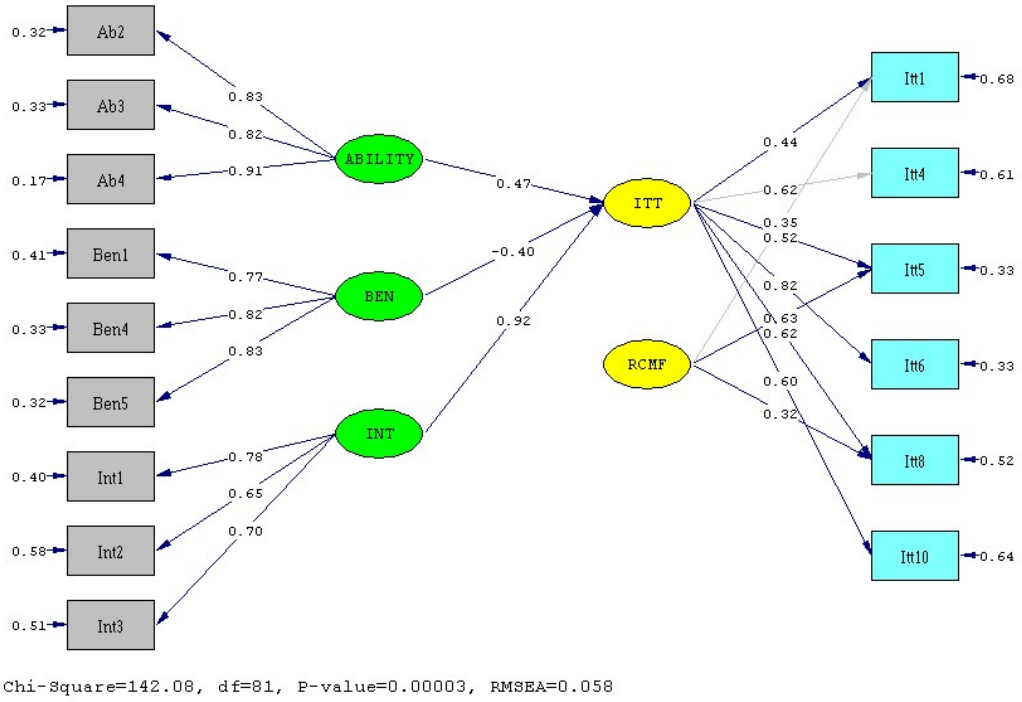


Chi-Square=38.82, df=24, P-value=0.02852, RMSEA=0.053



Chi-Square=9.36, df=6, P-value=0.15409, RMSEA=0.050

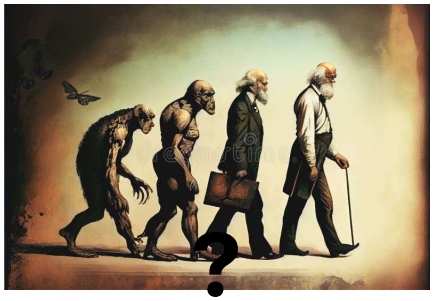
Test the Theory (Schoorman, Mayer, & Davis, 1996)



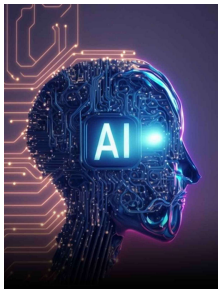
Chi-Square=142.08, df=81, P-value=0.00003, RMSEA=0.058

$$t = \frac{\bar{X}_A - \bar{X}_B}{\frac{S}{\sqrt{n}}}$$

Interim Research Finding:



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The Manual versus SurveyLM/AI collected data items are NOT equivalent.

What to do about it?

Technical Approach: Look at the SurveyLM Data responses:

1. **Use a different LLM model**
2. **Change the question(s) wording**
3. **Change the agent definitions**
4. **Change settings on the gpt-4o-mini model**
    - **Change the Justification**
    - **Critic**
    - **Analysis suggestions**
    - **Sampling and reasoning**
    - **Optional controls**

Possible Explanation for Findings?



Katie McDonnell (2026, p. 210) **Understanding Cultural Variation in AI-human Value Alignment**

Study systematically compares human respondents (WVS) and AI-generated counterparts across thematic domains of values.
"Collectively, the empirical findings demonstrate that **AI–human alignment is neither uniform nor randomly distributed**.
Instead, **alignment is systematically patterned across demographic, socioeconomic, cultural, ideological, and thematic dimensions**.
Differences in representational fidelity emerge across social groups, value domains, and broader cultural environments, indicating that alignment is shaped by the **same social and contextual structures** that organise **variation in human values** themselves.
These findings challenge interpretations of alignment as a purely technical problem and instead position it as a socially situated and culturally contingent phenomenon.

Thank you

Questions and Comments

Welcome

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